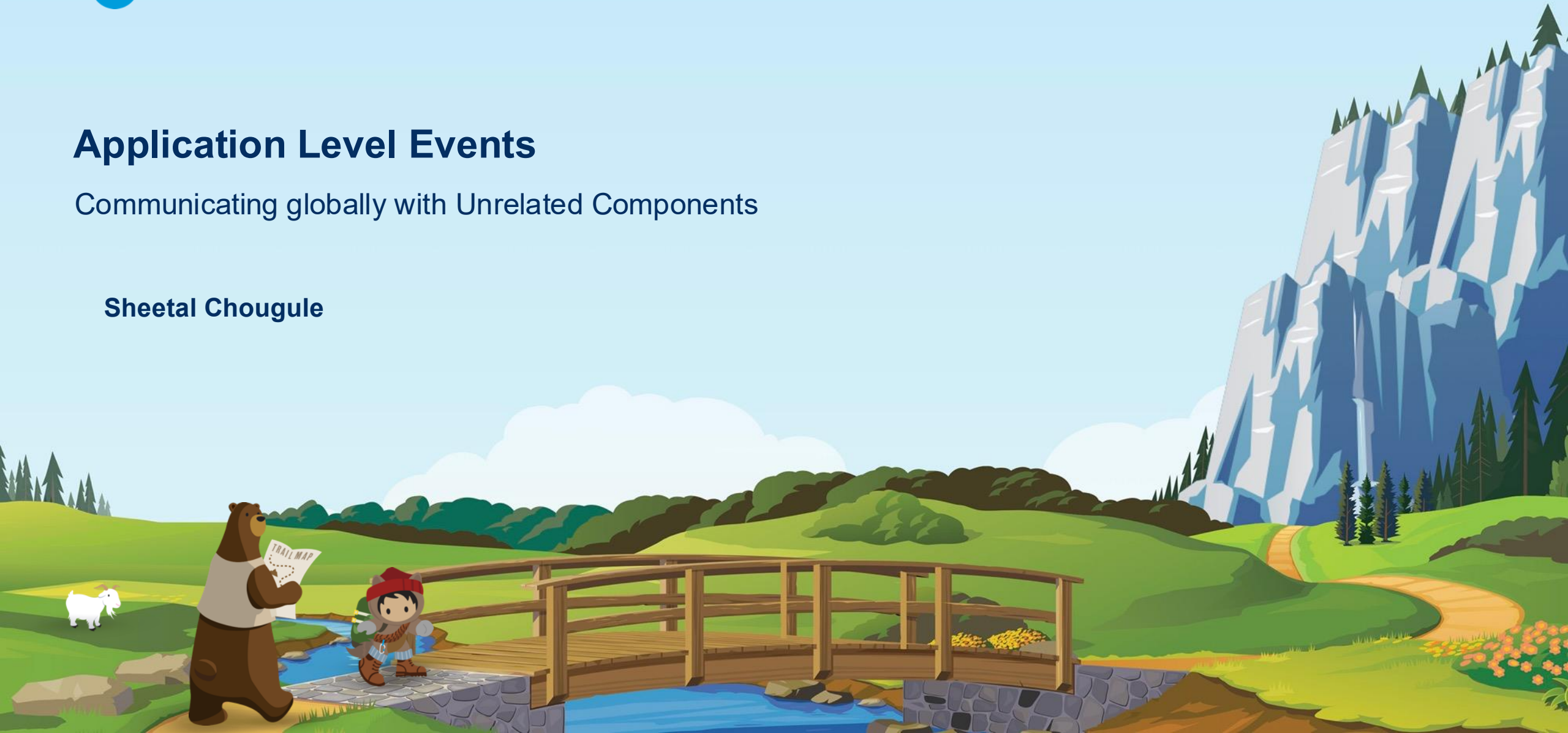


Application Level Events

Communicating globally with Unrelated Components

Sheetal Chougule

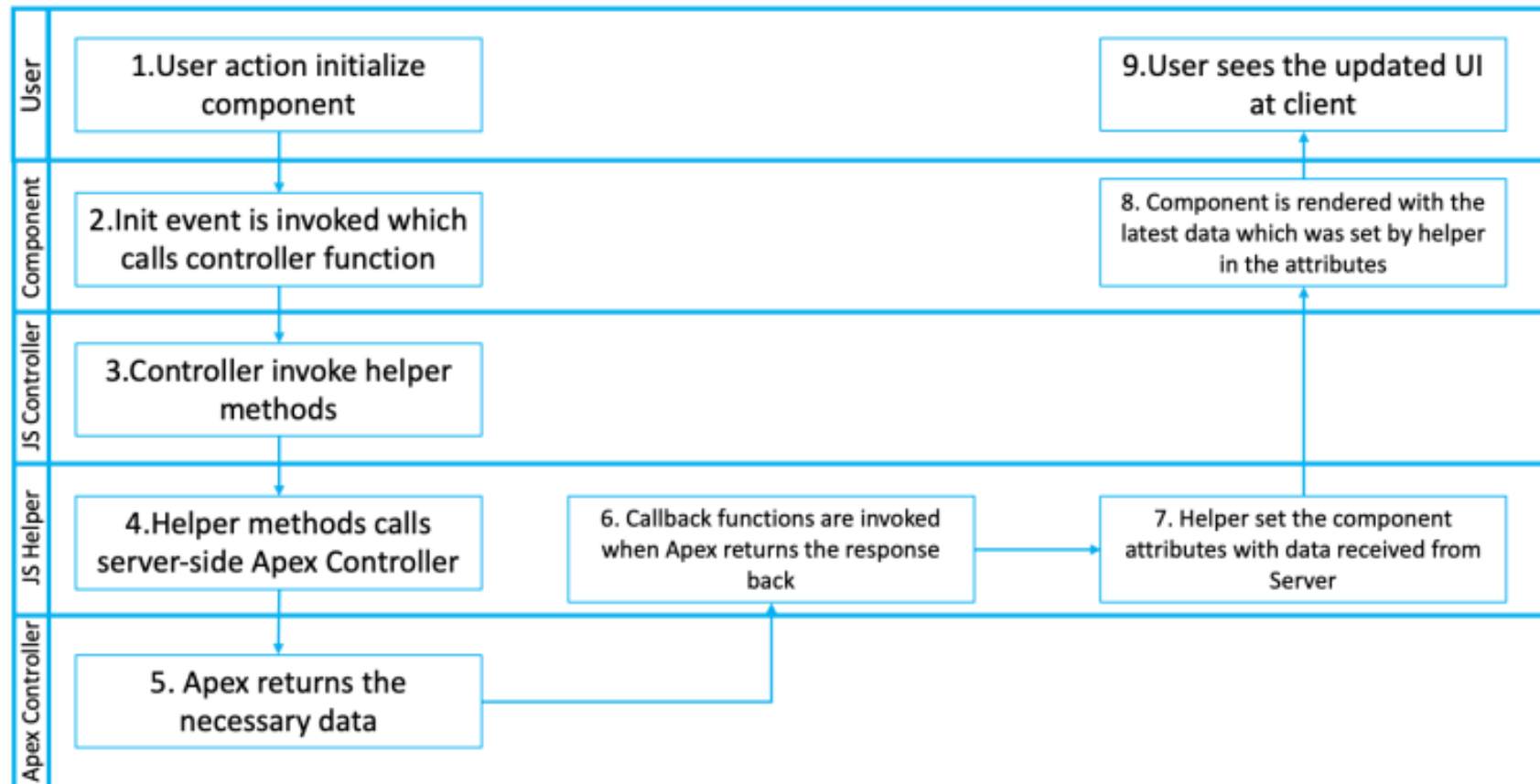




Mastering Application Events in Aura



Aura Component Lifecycle



What are Application Events?

- An Application Event is a type of event that is handled at the application level and can be broadcast across components that are not directly related in the component hierarchy.
- Unlike Component Events (which bubble up), Application Events propagate across the app.
- Any component can register to handle it, regardless of relationship.
- They follow a publish-subscribe model where a component fires an event, and any other component handling that event can receive the data.
- There are 4 major Steps:
 - Step 1:** Create Aura Application Event.
 - Step 2:** Register Event in Source/Sending component.
 - Step 3:** Fire Event from Source Component.
 - Step 4:** Handle the Event in Receiving Component.



When to Use Application Events?

- To trigger actions that affect the entire application, or when you need to communicate between unrelated components or those not directly connected in the component hierarchy.
- Suitable for actions like initializing data at the application start, handling errors, or performing cleanup tasks when the application ends.
- When you want to decouple components and avoid tight dependencies, application events are ideal.
- If a change in one component needs to be reflected in multiple other components, regardless of their location in the component hierarchy, application events are the right choice.

Use Case	Use Application Event?
Communication between sibling components	✓ Yes
Deep component hierarchy	✓ Yes
Parent-child interaction	✗ Prefer Component Event
Global triggers like logout, refresh	✓ Yes

Key Characteristics of Application Events

Global Scope

not tied to a specific component hierarchy. Any component can fire the event, and any component can listen for and handle it, provided they register to listen for that specific event type.

Publish-subscribe model

Components act as publishers (firing events) and subscribers (handling events).

Event propagation

Application events propagate through the application, meaning they are not limited to immediate parent-child relationships.

Component Terminology

- **Containment Hierarchy:** refers to a tree of components that has a top-level container as its root component. When an event is triggered, all components within the containment hierarchy are notified
- **Component Event Propagation:** How an event is propagated in the containment hierarchy depends on the phase that is configured in the event handler. There are two types of phases available for component events, namely, Bubble and Capture
 - **Bubble Phase:** The default phase used by a component event is the bubble phase, which behaves in a bottom-up manner. When an event is fired, 1) source component handles the event first. 2) parent component, 3) grandparent, and then each remaining component all the way up until the root component
 - **Capture Phase:** The order of the event propagation in the capture phase behaves in a top-down manner. 1) root component gets to handle the event first, 2) and the propagation traverses down the containment hierarchy until it reaches the source component
- **Stopping Event Propagation:** Event propagation can be terminated any time the stopPropagation method is executed in an event handler



Demo

Component Level Event – Creating
Browser Tiles

salesforce



Use Case



In a Training Institute app, an Admin page displays a dashboard to manage trainee enrollments.

When a trainee is selected in <traineeSearch>, we want to update <enrollmentSummary> to show their enrolled courses and allow modification.

Since the components are siblings or in separate containers, we use an Application Event.



Component Overview



Trainee Search

File	Purpose
traineeSelectedAppEvent.evt	Declares the custom event and its attributes for payload.
traineeSearch.cmp	Registers the custom event and then fires the event.
enrollmentSummary.cmp	Handles the event raised by the traineeTile.

Project Structure



```
aura/  
├── traineeSearch/  
│   ├── traineeSearch.cmp  
│   ├── traineeSearchController.js  
│   └── traineeSearchHelper.js  
├── enrollmentSummary/  
│   ├── enrollmentSummary.cmp  
│   ├── enrollmentSummaryController.js  
│   └── enrollmentSummaryHelper.js  
events/  
└── traineeSelectedAppEvent.evt  
apex/  
├── TraineeController.cls  
└── EnrollmentController.cls
```



Component vs Application Events



When to use Component Events vs Application Events

Prefer component level events over Application level events

Component Events

Simple communication within a component or its immediate children/parent:

When you need a component to communicate with its immediate parent or children, a component event is usually the best choice.

Localized UI updates:

If you need to update a specific part of the UI based on an action in a child component, component events are appropriate.

Application Events

Wider scope:

Application events are for broader communication across the entire application, using a publish-subscribe model.

More complex setup:

They require more setup and are generally used for more global actions like navigation or data updates that need to be reflected throughout the application.

thank you

Family Reunion

